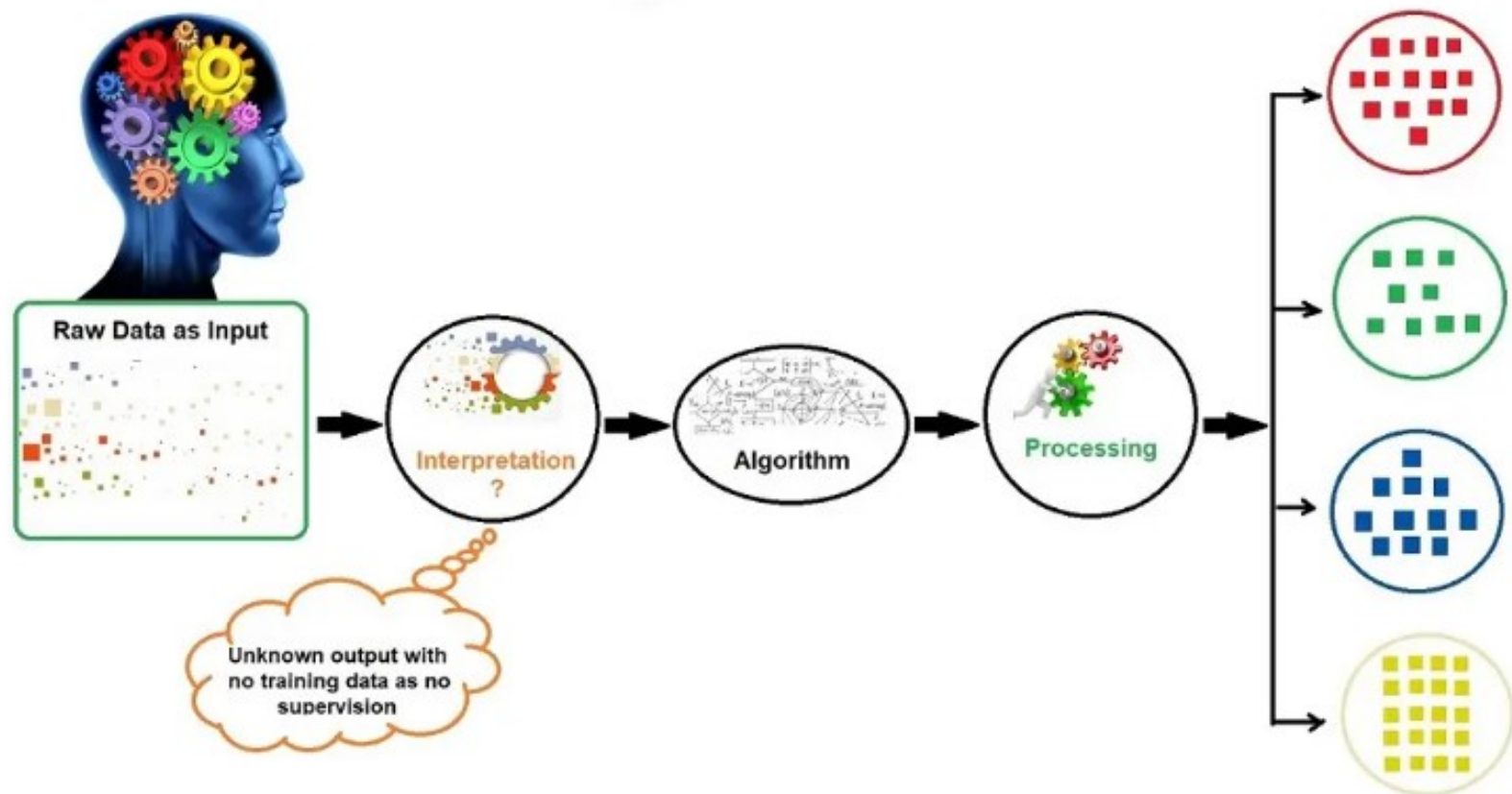


UNIT 2
Basic of Machine Learning



Machine Learning (ML) is a subset of Artificial Intelligence (AI) that allows computers to learn from data and make decisions without being explicitly programmed. Essentially, ML algorithms identify patterns in data and use those patterns to predict or classify new, similar data. This ability to learn from data and improve over time is what makes ML so powerful.

Applications of Machine learning

Machine learning is seamlessly integrated into our daily lives and used across diverse industries.

Natural Language Processing (NLP): Virtual assistants like Siri and Alexa, translation software like Google Translate, and chatbots all rely on NLP for speech recognition and language understanding.

Image Recognition: Social media platforms use ML for facial recognition and object detection to auto-tag friends or categorize images.

Self-Driving Cars: Autonomous vehicles utilize ML to interpret sensor data, recognize objects, and make real-time driving decisions.

Fraud Detection: Financial institutions use ML to identify suspicious transactions by detecting patterns in large datasets, helping to prevent financial losses.

Medical Diagnosis: ML analyzes patient data to assist doctors in diagnosing diseases, predicting patient outcomes, and developing personalized treatment plans.

Targeted Marketing: Analyzing customer behavior to segment customers and create personalized marketing campaigns.

Recommendation Systems: Services like Netflix and Spotify use ML to suggest movies or music based on user preferences and past behavior.

Machine learning is a transformative technology that continues to evolve, offering innovative solutions and impacting various sectors from business to healthcare, to entertainment.

Processes involved in Machine Learning

The machine learning lifecycle is a systematic process of developing and deploying effective ML models, consisting of several stages.

Problem Definition: Clearly defining the problem that needs to be solved and the objectives to be achieved.

Data Collection and Preparation: Gathering, cleaning, and labeling the data, which is crucial for building and training an accurate model.

Model Selection and Engineering: Choosing the right algorithms and building the model architecture based on the problem and data type.

Training and Validation: Training the model with the prepared data and validating its performance on unseen data to avoid overfitting and underfitting.

Evaluation and Testing: Rigorously evaluating the model's performance on test data using relevant metrics and making improvements.

Deployment and Integration: Deploying the model into a live environment, making its predictions accessible through APIs, web apps, etc.

Monitoring and Maintenance: Continuously monitoring the model's performance and retraining it with new data to ensure continued effectiveness.

Types of Machine learning

Machine learning algorithms are broadly categorized into three types based on how they learn and the type of data they use.

1. Supervised Learning:

Description: Algorithms are trained on labeled data, where each input has a known output, learning the relationship between them.

Purpose: Predicting a categorical label (classification) or a continuous value (regression) for new, unseen data.

Examples: Email spam detection, image recognition, medical diagnosis based on patient data.

2. Unsupervised Learning:

Description: Algorithms work with unlabeled data to identify patterns, groups, or structures on their own, without predefined output variables.

Purpose: Discovering hidden patterns, grouping similar data points (clustering), or reducing the number of features (dimensionality reduction).

Examples: Customer segmentation based on purchasing behavior, anomaly detection for fraud prevention, recommendation systems based on user preferences.

3. Reinforcement Learning

Description: An agent learns by interacting with an environment, receiving rewards for good actions and penalties for bad ones, and adjusting its behavior to maximize rewards over time.

Purpose: Optimal decision-making in dynamic environments where the agent's actions affect future states.

Examples: Training robots, autonomous vehicles, game playing, personalized training systems

Applications

Real life examples

Recommendation Systems

Netflix suggesting movies and shows based on viewing history, Amazon recommending products based on past purchases, and Spotify curating personalized playlists.

Natural Language Processing (NLP)

Voice assistants like Siri and Google Assistant responding to commands, translation tools like Google Translate translating between languages, and chatbots providing customer support.

Fraud Detection

Banks using ML models to flag suspicious transactions, and credit card companies using ML to identify potentially fraudulent purchases.

Autonomous Vehicles

Self-driving cars navigating streets and making driving decisions, and ride-sharing apps like Uber and Lyft optimizing routes and pricing.

Machine Learning & Applications